ANALYTICAL CUMULATIVE EXAM

September 21, 1991

Arguably the most important paper in cosmochemistry this year was published on pages 2317-2337 of the August issue of Geochimica Cosmochimica Acta. It was written by R.N. Clayton, T.K. Mayeda, J.N. Goswami and E.J. Olsen, and was entitled "Oxygen isotope studies of ordinary chondrites". The oxygen region of a Table of Nuclides is attached.

(20) 1. Oxygen 16 is synthesized in the interiors of $>1M_O$ stars by helium burning, in particular by alpha capture on carbon 12. Write the complete nuclear reaction and calculate the energy released per reaction.

(20) 2. If the reaction cross section for alpha capture on ^{12}C is σ, and the flux of alpha particles was ϕ $cm^{-2}s^{-1}$, derive an expression for the rate of production of ^{16}O in the stellar interior.

(20) 3. Assuming that $\sigma = 200mb$, and $\phi = 10^{24}cm^{-2}s^{-1}$, what is the rate of energy production per mole of target atoms in the star.

(20) 4. In most situations, including most laboratory experiments, the three isotopes of oxygen are fractionated according to their masses, the ratio of ^{17}O to ^{16}O varying half as much during a physical/chemical process as the ratio of ^{18}O to ^{16}O. What is this fractionation process termed, and what is the significance of Clayton et al.'s discovery that in meteorites the $^{17}O/^{16}O$ ratio is directly proportional to the $^{18}O/^{16}O$ ratio.

(20) 5. A popular method of oxygen 16 analysis is the use of a (n,p) reaction using 14 MeV neutrons and measuring the activity of the product nuclide. What is the product nuclide and its half-life?